# UAE Public Service Continuity Agent

A deterministic, policy-grounded prototype for synthetic public-service continuity evaluation

| Submission field | Value | Evidence status |
|---|---|---|
| Team | YiYuan Qiming | OWNER-SUPPLIED |
| Leader / members | 周玉晨 (Yuchen Zhou) / Single-member team | OWNER-SUPPLIED |
| Scenario | Synthetic extreme-heat continuity testbed | S1-S5 OWNER APPROVAL NOT VERIFIED |
| Repository | No remote URL asserted | REVIEWER ACCESS NOT VERIFIED |

## 1. Agentic AI Solution and Use Case

This project is a deterministic local prototype for synthetic public-service continuity evaluation during an extreme-heat event. Event Intelligence, Policy Adviser, Simulated Coordinator and Authorization/Audit roles process staged events and produce source-cited, side-effect-free recommendations.

The roles are ordinary functions in one Node.js process. They are not a generative autonomous multi-Agent system. No external AI Provider, government system, real user data or institutional authority is used.

The original S1-S5 fixture remains **WORKING_ASSUMPTION_NOT_OWNER_APPROVED**. The rescue package also contains a separate T1-T3 **TECHNICAL_TEST_SCENARIO** with **OWNER_APPROVAL=NOT_GRANTED**.

## 2. Use-Case Requirements

The prototype must preserve event identity and ordering, keep incomplete or conflicting evidence visible, retrieve only admitted policy claims, cite source and locator, reject unsupported source claims, fail closed on uncertain authorization, preserve an audit chain and keep human review explicit.

Power restoration is not service restoration. The prototype does not reopen a service without separate evidence for facility, service, capacity, demand and human acceptance.

# 3. Architecture and Trust Boundaries

| Role | Implemented behavior | Boundary |
|---|---|---|
| Event Intelligence | Validates identity, source label, domain sequence, gaps, lateness and current facts | Synthetic event logic only |
| Policy Adviser | Retrieves admitted claims by exact metadata/tag overlap and emits citations | No vector retrieval; no legal conclusion |
| Simulated Coordinator | Produces side-effect-free continuity proposals | No institutional authority |
| Authorization/Audit | Checks action state, evidence and fencing token semantics | Simulation-only; no external executor |

## Evidence Chain

Each audit record contains a canonicalized payload, prior-record hash and own SHA-256. Event records bind identity, source, observed time, payload and hashes. A manifest binds run, audit and input hashes. This detects local inconsistency; it is not immutable storage, external timestamping or a digital signature.

## Verifier Boundary

Execution does not load the Oracle. The separate verifier checks run artifacts, input bindings, Agent/audit consistency, state semantics, unsafe actions and token rules. It is repository-local machine logic sharing the same operator, not independent human acceptance or third-party certification.

## 4. Evaluation Safety Rule

Unknown sources, uncertain state, missing service evidence or missing capacity evidence cause a fail-closed decision. Fencing tokens are stage/action scoped, expiring and single-use. Every effect in this candidate is marked simulation-only.

# 5. Policy Knowledge Base and Agent Execution

The metadata-first knowledge base registers 12 public source records, admits 8 scoped claims and counts 7 supporting UAE sources. Each source record includes issuer, title, official URL, retrieval date, recorded byte size, SHA-256 field, scope, license and redistribution disposition. Raw UAE/ITU snapshots are excluded because source-specific redistribution rights are not uniformly verified.

| Capability | Current result | Status |
|---|---|---|
| Retrieval | Exact metadata/tag overlap | IMPLEMENTED_AND_VERIFIED |
| Policy Adviser query | 8 citations; scoped answer; authority gap; no legal conclusion | PASS |
| Source-byte verification | Not performed in public metadata mode | CONDITIONAL |
| Vectorization | No embeddings or vector store | NOT_IMPLEMENTED |

A real question was executed against the rescue copy: what privacy, oversight and resilience controls apply to a synthetic continuity prototype, and does the admitted source set establish authority to reopen after power returns? The response cited eight claims, stated that the source set provides scoped governance guidance only, recorded the missing heat/public-cooling mandate, returned **legal_conclusion=false** and set **human_review_status=REQUIRED_NOT_PERFORMED**.

This is a deterministic function over metadata-only claims. It does not expose hidden chain-of-thought, does not call an external model and does not provide legal advice.

# 6. Technical and Policy Evaluation Scenarios

| Scenario | Stages and behavior | Result |
|---|---|---|
| Original S1-S5 | Baseline, power interruption, route gap, unknown capacity source and power restoration without service evidence | 25/25 verifier assertions; scenario approval not verified |
| New T1-T3 | Initial event, new power evidence, then authority/policy conflict with unknown source | 8/8 local assertions; Owner approval not granted |

The new fixture is explicitly technical and does not rewrite the original scenario's authorization state. In T2 the service-availability request is blocked after power interruption. In T3 power restoration and demand surge arrive with an unknown authority claim; the claim is rejected, capacity remains unknown and reopening is blocked.

The rescue execution passed 20/20 tests. Existing S1-S5 local metrics remain descriptive: 4/4 unsafe requests blocked, 1/1 unknown claim rejected and 0/1 false recovery observed. All metric thresholds remain **PROPOSED_NOT_VALIDATED**.

Visible audit fields include event ID, timestamp, scenario stage, input reference, retrieved source IDs, policy reference, Agent output, decision status and human-review status. The S1-S5 chain contains 57 records; T1-T3 contains 30 records.

## 7. Failure and Recovery Boundaries

S5 restores only the power fact. It does not infer service or capacity recovery. Stale evidence, replayed or revoked tokens, source mismatch, audit tampering and false recovery outputs are covered by the local tests and verifier.

## 8. Required Mappings, Gaps and Submission Boundary

| Requirement | Local evidence | Status |
|---|---|---|
| ITU-T Y.3172 | Engineering architecture mapping | DOCUMENTED_ONLY; not conformity |
| ITU AI Readiness Framework 2.0 | Scoped readiness mapping | DOCUMENTED_ONLY; not national assessment |
| Local policy mapping | Data protection, digital government, AI charter, adoption, cyber and responsibilities | PARTIAL |
| Policy/standards gaps | Authority, corpus, licensing, freshness and global-case gaps | DISCLOSED |
| GitHub repository | Local essential-code package only | NOT VERIFIED |

Material gaps are explicit: no verified heat/public-cooling operating mandate, no complete domain corpus, no global comparative policy-case corpus, no vector or distributed knowledge base, no ITU Lab hosting, no production or institutional validation, and no remote reviewer-accessible repository.

The public candidate remains **UNLICENSED / ALL RIGHTS RESERVED**. It includes only event-scoped code, synthetic fixtures, source metadata, scoped paraphrases, links and evidence. Raw source snapshots, internal governance records, credentials, personal data and unrelated YiYuan assets are excluded.

## 9. Final Status

| Field | Status |
|---|---|
| Technical report | PASS locally; five pages |
| Knowledge base | CONDITIONAL; metadata-only and incomplete |
| Agent execution | PASS locally |
| Scenario test | PASS locally; authorization not granted |
| Auditability | CONDITIONAL; local evidence only |
| Official submission route | NOT_VERIFIED |
| Final submission | NO_GO |

This report is a local submission candidate, not an official acceptance or submission receipt. Final external action requires documented Owner/organizer resolution of repository access and scenario authorization. No public repository, email, form or upload was created by this rescue run.